LLM GATEWAY CACHE · PARAPHRASE BENCHMARK · 2 LANGUAGES · 9,917 QUESTION PAIRS

Can a cache answer in the model's place without ever answering the wrong question?

We put a five-layer cache in front of a language model and replayed 9,917 question pairs that people had already labelled as “same question” or “different question”. Every time the cache answered from memory instead of calling the model, we checked whether it was allowed to. This page is the short answer.

TURN ON

Exact match

Tidies the wording, then looks the question up by a fingerprint of its text.

0 wrong answers

It only fires when two questions are letter-for-letter the same after tidying, so there is nothing for it to get wrong. Free, and safe in both languages.

ENGLISH ONLY

Keyword match

Finds an earlier question that uses the same important words.

6% wrong in English, **33%** in Persian

One wrong answer in sixteen is a tolerable price in English. One in three is not, so in Persian this layer has to be paired with a second check.

NOT SAFE IN PERSIAN

Meaning match

Compares what questions *mean*, so it catches rewordings that share no words.

15% vs **63%** wrong

The strongest layer in English and the most dangerous one in Persian: nearly two of every three Persian answers it reused belonged to a different question. Worse than not caching at all.

PERSIAN ONLY

The guard

Deletes whatever two questions have in common and judges only the part left over.

0.88 : 1 vs **6.5 : 1**

Right answers given up for each wrong one stopped. A fair trade in Persian, a bad one in English, where it throws away six and a half good answers to stop one bad one.

How to read the numbers. Percentages are shares of the cache's own decisions, not of all traffic. The one that matters is how often a reused answer was the right one; with these sample sizes a gap smaller than about ±3.9 points is inside chance, so anything smaller is drawn faded and described as no change.

1 · HOW WE TESTED

Store the first question, ask the second, see if the cache answers

1 Empty the cache and store question A with its answer → **2** Ask question B → **3** Note whether the cache answered, and which layer did → **4** Compare with the human label for the pair

The cache starts empty for every pair, so an answer can only have come from question A. If the pair is labelled *same question*, answering from memory is right. If it is labelled *different question*, answering from memory means a person was handed a reply to a question they never asked — the failure this whole exercise is about. No model was called: the pairs already carry human labels, so the cache's decision can be scored on its own. The full run took 30 minutes and costs nothing to repeat.

	SET — WHAT ONE PAIR LOOKS LIKE	MADE OF	PAIRS	SAME QUESTION
1	**MSRP (English)** “The deal was announced Monday.” vs “Monday brought news of the deal.”	news sentences	5,301	67%
2	**ParsiNLU (Persian)** «پایتخت فرانسه کجاست؟» در برابر «پایتخت ایران کجاست؟»	search questions	4,616	40%

How often it was right. Of all the times the cache answered from memory, the share where that answer really did belong to the question being asked. This is the number to judge a cache by: a cache that answers everything from memory looks busy and is useless.

How much it caught. Of all the pairs where reuse *was* allowed, the share the cache actually spotted. This is the saving. Pushing it up while the first number falls does not save money; it produces wrong answers cheaply.

What the request falls through

`cheapest` **Exact** same text after tidying → **Keyword** same important words → **Meaning** same sense, other words → **Examples** near misses shown to the model → `dearest` **The model** a full call

Each step costs more than the one before it — a fingerprint lookup, then an index search, then a meaning comparison, then a full call — so the expensive steps only happen once the cheap ones have failed. The first step confident enough to answer stops the fall. The guard sits across the middle two, the only ones that ever guess.

Both sets are deliberately hard: the pairs labelled *different* use almost as many of the same words as the pairs labelled *same*, so counting shared words cannot separate them. Meaning is compared with `cached:sbert`, a small multilingual model that reads both scripts.

One dial sets everything, and it has to be set per language

The dial is how alike two questions must look before the cache dares reuse an answer. Turning it up buys accuracy with savings. Each cell is a share; blue means it clears the bar we set for that column (right at least 85% of the time, catching at least 30%), red means it fails it.

clears the bar · fails the bar · in between · not run

How alike, at least	English share right	English share caught	Persian share right	Persian share caught
0.60 — loosest	77% in between	42% clears it	36% fails it	41% clears it
0.70	81% in between	36% clears it	37% fails it	41% clears it
0.80	86% clears it	24% in between	40% fails it	37% clears it
0.85	88% clears it	17% in between	44% fails it	31% clears it
0.90	90% clears it	10% fails it	52% fails it	22% in between
0.95 — strictest	95% clears it	4% fails it	70% in between	8% fails it

English, share right climbs steadily and clears the bar from 0.80 upwards.

English, share caught is what pays for it: 42% at the loosest setting, 4% at the strictest.

Persian, share right never clears the bar — not at any setting on this dial.

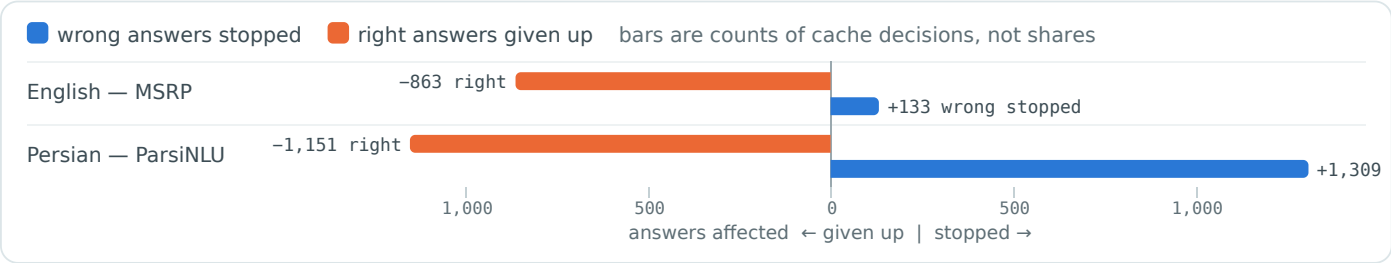
Persian, share caught collapses just as fast, so there is no setting that is both safe and worth having.

There is no blue cell anywhere in the two Persian columns. That is the finding, and no amount of tuning changes it.

3 · THE GUARD

The guard does not make reuse more accurate; it makes reuse rarer

Many prompts share a long fixed preamble — a system instruction, a set of worked examples, a block of retrieved context. That shared text dominates any similarity score, so two prompts that differ only where the answer is decided still look 80% alike. Raising the dial does not help, because the score is inflated for everything. The guard deletes what the two prompts have in common and judges only the leftover. Below is everything it removed, counted.



English — a bad trade

6.5 : 1

863 right answers given up to stop 133 wrong ones · reuse fell 1,405→409

Persian — a fair trade

0.88 : 1

1,151 given up, 1,309 stopped · reuse fell 3,229→769

Neither accuracy gain is big enough to believe. Reuse got +2.0 points more accurate in English, against a margin of error of ±3.4, and +3.8 in Persian against ±3.9. Both sit inside the noise, so the guard does not measurably raise the *rate* at which reuse is right. What it does change, and by a lot, is the *number* of answers reused — and therefore the number of wrong ones that reach a person.

One word changed, and it is the word that decides the answer

ALREADY CACHED	What is the capital of France?
NOW ASKED	What is the capital of Iran?
WHAT DIFFERS	france vs iran

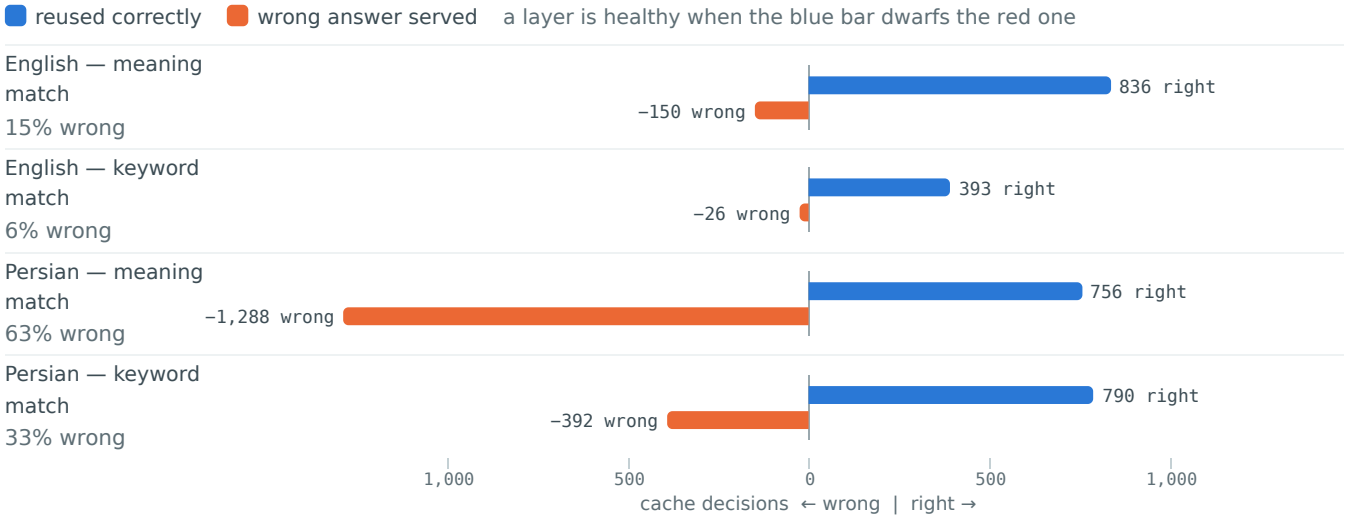
WITHOUT THE GUARD	WITH THE GUARD	RIGHT ANSWER
Paris	asks the model	Tehran

Shared text 83% · guard said: **blocked** — template dominates (83%); the parts that differ only match 0%. Numbers and names are checked before anything else.

4 · WHICH LAYER GETS IT WRONG

One layer is responsible for almost all the damage

With the guard switched off, so each layer is seen as it really behaves. The keyword layer is the cautious one in both languages. The meaning-match layer is where the risk lives, and in Persian it is upside down: it hands out more wrong answers than right ones. Exact matching appears in neither chart because it made no mistakes to plot.



Same sentence, different number

ALREADY CACHED	Convert 100 kilometres to miles.
NOW ASKED	Convert 250 kilometres to miles.
WHAT DIFFERS	100 vs 250

WITHOUT THE GUARD	WITH THE GUARD	RIGHT ANSWER
62.1 miles	asks the model	155.3 miles

Shared text 80% · guard said: **blocked** — different numbers (100.0, 250.0). A one-character edit that changes the whole answer.

Different words, same question — this one must still be reused

ALREADY CACHED	How do I restart a Docker container safely?
NOW ASKED	How can I reboot a containerized application without downtime?
WHAT DIFFERS	do i restart a docker container safely vs can i reboot a containerized application without downtime

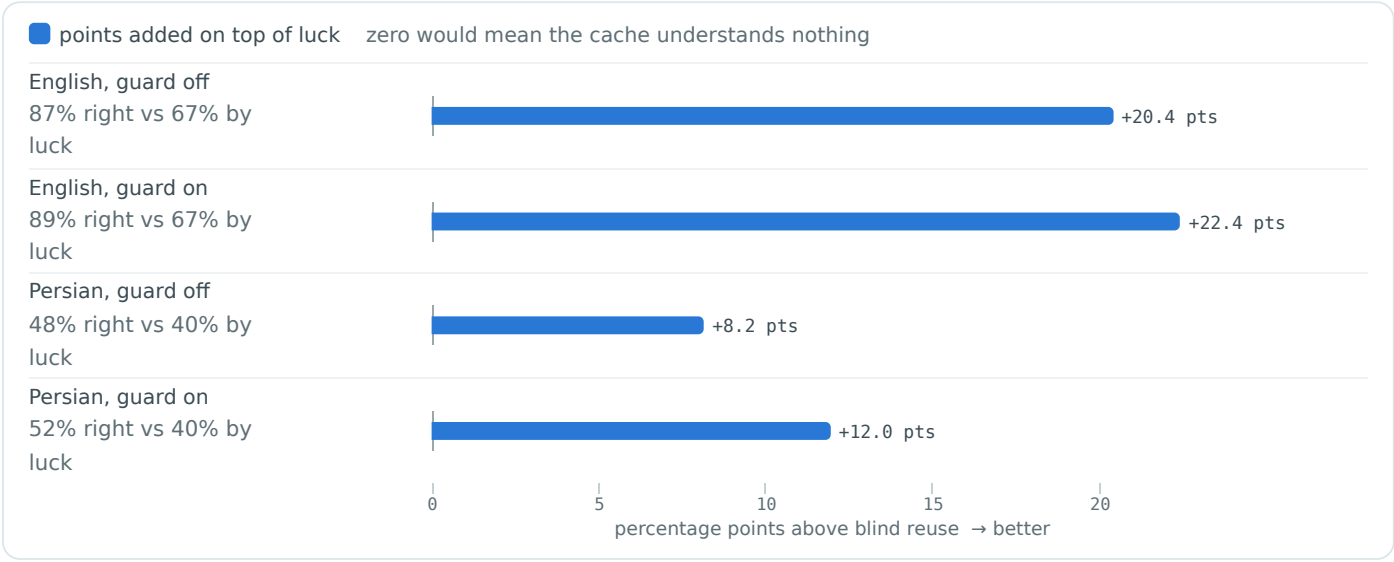
WITHOUT THE GUARD	WITH THE GUARD	RIGHT ANSWER
docker restart <name>, or drain first	docker restart <name>, or drain first	docker restart <name>, or drain first

Shared text 11% · guard said: **reused by both** — no dominant shared block. Only 11% of the words are shared, so the guard stays out of the way and the saving is kept.

5 · ENGLISH AGAINST PERSIAN

The language gap is real, but it is 12 points, not 40

English reuse was right 87% of the time, Persian only 48% — a gap of 40 points. But the two sets are not built alike: 67% of the English pairs really are the same question, against 40% of the Persian ones. A cache that reused blindly would therefore already score 67% in English and only 40% in Persian, without understanding anything. What the cache adds on top of that luck is the fair comparison.



Gap at face value

40 pts

before correcting for how the two sets were built

Gap once corrected

12 pts

Persian adds 8 points on top of luck where English adds 20 — about 40% as much

The same trap in Persian

ALREADY CACHED	پایتخت فرانسه کجاست؟
NOW ASKED	پایتخت ایران کجاست؟
WHAT DIFFERS	فرانسه vs ایران

WITHOUT THE GUARD	WITH THE GUARD	RIGHT ANSWER
پاریس	asks the model	تهران

Shared text 67% · guard said: **blocked** — template dominates (67%); the parts that differ only match 0%. The guard works on Persian without knowing any Persian: it compares only what is left over.

A real Persian paraphrase, which must survive the guard

ALREADY CACHED	چطور یک کانتینر داکر را با خیال راحت ری‌استارت کنم؟
NOW ASKED	برای راه‌اندازی مجدد کانتینر داکر بدون قطعی چه کنم؟
WHAT DIFFERS	چطور یک کانتینر داکر را با خیال راحت ری استارت vs برای راه اندازی مجدد کانتینر داکر بدون قطعی چه

WITHOUT THE GUARD	WITH THE GUARD	RIGHT ANSWER
`docker restart <name>`	`docker restart <name>`	`docker restart <name>`

Shared text 9% · guard said: **reused by both** — no dominant shared block. And it lets a real Persian rewording through.

Recommendations

1. **In Persian, switch meaning-match off and keep exact and keyword matching.** The meaning-match layer was wrong 63% of the time (1,288 of 2,044 reuses), and the strictest setting on the dial still only reaches 70% — while catching 8%, which is not worth having. Exact matching is safe by construction and costs nothing.
2. **In English, set the dial by what a wrong answer costs you, and leave the guard off.** 0.85 gives 88% right at 17% caught; 0.95 gives 95% at 4%. The guard costs 6.5 right answers per wrong one stopped here, which the dial does more cheaply.
3. **Turn the guard on wherever shared templates are common, or wherever reuse is already unreliable.** A filter earns its cost only when what it filters is mostly bad. English reuse was 87% right before the guard, so most of what it threw away was right; Persian reuse was 48% right, so most of what it threw away was wrong.
4. **Measure on your own traffic before trusting any of these numbers.** This benchmark is built entirely from look-alike pairs and contains almost no repeats, so it measures the hard case only. Real traffic repeats itself, where exact matching does the work for free.

Found along the way: the meaning-match layer was quietly using an English-only model. The default sentence model has no Persian word pieces at all, so it mapped unrelated Persian sentences to almost the same point and the cache confidently served the wrong answers. The model is now configurable and defaults to a multilingual one, and the guard was added as a second, language-independent line of defence — it compares only the leftover text, so it catches the Persian trap without knowing Persian. The code is pinned by 129 automated tests.

Limits of this test

- Both sets are adversarial by construction. Non-matching pairs were chosen to resemble matching ones, so the true hit rate on ordinary traffic will be far higher and so will accuracy.
- The two sets have different proportions of matching pairs, so their accuracy figures are not directly comparable without the correction on page 5.
- Only one sentence model was tested, so the Persian result may belong to that model rather than to Persian. A larger multilingual model would separate those two explanations.
- 1 Persian pair was labelled “different” while both sides were letter-for-letter identical. It was detected and excluded; others may remain, which would make the Persian numbers look worse than they are.
- Speed and money were not measured here — this test judges correctness only. Per-model latency and token use are measured separately against the live endpoint.

About this report. The text and layout of this report were written with the help of Claude (an AI assistant). The experiments themselves were really run: 9,917 labelled question pairs replayed through the cache twice, with the guard off and on, plus a six-point sweep of the similarity dial — 30 minutes of compute, no model calls. Every figure on these pages is parsed straight out of that run's own output file by the script that generated this HTML, and every example was produced by calling the guard live while the page was built.